

Chapter 59 — Differences of Sexual Development: Etiology, Evaluation, and Medical Management

- **Differences (disorders) of sex development (DSD):** conditions in which chromosomal, gonadal or anatomical sex characteristics are not typically binary male or female. Familiar examples are **congenital adrenal hyperplasia (CAH)**, **ovotesticular DSD (OVT DSD)** and **androgen insensitivity syndrome (AIS)**.
- Care is **multispecialty** and constantly evolving; controversies common to many DSDs (sex designation, surgical timing, gonadal management) are treated together at the end of the chapter.

TERMINOLOGY

- Nomenclature has evolved "**hermaphrodite**" → "**intersex**" → "**DSD**". The landmark **2006 consensus statement (Lee et al.)**, from patients, clinicians and advocates, replaced "intersex" with "DSD" and retired "sex reversal" and "hermaphrodite" as pejorative (Table 59.1).
- Many affected people find "disorders" pathologising and prefer "**intersex**" (especially as identity), "**differences of sex development**", or **their diagnosis**. There is no consensus; surveys show patients and parents **most often name themselves by diagnosis**.
- "**Atypical**" or "**nonbinary**" **genitalia** may be more accurate and less stigmatising than "ambiguous genitalia"; the chapter uses "DSD" (differences) and "atypical genitalia". Genital photography in publications is controversial, so photographs are limited to those of greatest educational value.

TABLE 59.1 REVISED NOMENCLATURE

Previous term	Proposed term (2006)
Intersex	DSD
Male pseudohermaphrodite	46,XY DSD
Female pseudohermaphrodite	46,XX DSD
True hermaphrodite	Ovotesticular DSD
Mixed gonadal dysgenesis	Mixed gonadal dysgenesis (unchanged)
XX male or XX sex reversal	46,XX testicular DSD
XY sex reversal	46,XY complete gonadal dysgenesis

From Lee PA, Houk CP, Ahmed SF, Hughes IA; International Consensus Conference on Intersex, Pediatrics 118:e488–e500, 2006, Table 1.

- **Pattern:** conditions are now named **by karyotype first**; only mixed gonadal dysgenesis is unchanged. **Trap:** "XY sex reversal" is **complete gonadal dysgenesis**, not androgen insensitivity; "true hermaphrodite" is **OVT DSD**.

What Belongs Under the DSD Umbrella

- A broad umbrella groups clinically similar diagnoses for research, connects people with similar manifestations for support, and reflects shared need for multidisciplinary expertise. Membership of **proximal hypospadias, CAH and sex chromosome conditions** is disputed.
- **Proximal hypospadias** (penoscrotal, scrotal, perineal): some call all cases "46,XY DSD of unclear etiology"; others reserve "DSD" for a named diagnosis, avoiding social and legal controversy. Because a named diagnosis has many benefits (Table 59.2), **offer endocrine and genetic testing for proximal hypospadias, even with descended testes**.
- **CAH** is called the commonest 46,XX DSD, yet many affected people **do not identify as DSD or intersex**: they regard it as an **endocrine condition**, and **most 46,XY individuals with CAH have typical genitalia**.

They most often describe themselves by "just the diagnosis", which can deter attendance at "DSD" clinics or studies.

- **Turner (TS) and Klinefelter (KS) syndromes** are within the 2006 scope, but TS without Y and KS carry **no increased gonadal malignancy risk** and rarely have atypical genitalia, so they are often managed elsewhere; urologists still need their gonadal and systemic features.

TABLE 59.2 POTENTIAL IMPLICATIONS OF A DSD DIAGNOSIS

Topic	Implication
Sex designation / future gender identity	Informs infant designation from adult gender identity data for that diagnosis, where available
Surgical counselling	Possibly higher hypospadias complication rates
Fertility potential	Predicts fertility by diagnosis; eligibility for experimental gonadal tissue cryopreservation
Hormonal function	Proactive monitoring for peer-concordant puberty, growth, bone health
Gonadal tumour risk	Anticipatory guidance, screening or gonadectomy where risk is known
Multisystem comorbidities	Targeted comorbidity screening
Genetic transmission	Counselling on future pregnancies and offspring
Psychosocial health	Peer support; behavioural health support; benefit of an explanation for the genital difference

From Johnson EK, Whitehead J, Cheng EY: Differences of sex development: current issues and controversies. Urol Clin North Am 50(3):433–446, 2023, Table 3.

- **Purpose:** the case for pursuing a named diagnosis — every domain becomes predictable once named.
Trap: assuming proximal hypospadias with descended testes needs no work-up.

Concept — *Consensus nomenclature is karyotype-based, but the people it describes often prefer diagnosis-specific terms. The boundary of "DSD" is drawn by shared features — atypical genitalia and gonadal tumour risk — not by karyotype alone, which is why CAH, TS and KS sit uneasily inside it and proximal hypospadias arguably belongs.*

TYPICAL SEX DEVELOPMENT

- Three sequential steps: **chromosomal sex → gonadal sex → phenotypic sex**. **Chromosomal sex** is fixed at fertilisation by the karyotype. **Gonadal sex** is the histology and function of the gonad, which starts **bipotential**; gene activation and suppression produce testis or ovary.
- **Primordial germ cells (PGCs)** migrate from the dorsal yolk-sac wall through the mesentery to the medioventral urogenital ridge, starting in **week 5** (Fig. 59.1), then become spermatogonia or oogonia as the testicular and ovarian cords differentiate.
- **SRY (sex-determining region Y)**, among the earliest testis-determining genes, is a transcription factor that supports cells of the male urogenital ridge and activates a cascade including **SOX9 upregulation**.
- These genes also act elsewhere — **MYRF** (heart, eye), **SOX9** (bone, cartilage), **WT1** (kidney), **SF1** and **DAX1** (adrenal) — which explains syndromic DSDs (Table 59.3).
- **Leydig cells** (interstitium) make testosterone; **Sertoli cells** (within tubules) support spermatogenesis. Testicular PGCs are **held out of meiosis until puberty**. **Testis development precedes ovarian by about 2 weeks**.
- **Ovary** (less understood) needs **Rspo1/Wnt4/β-catenin** and **Foxl2/Runx1**: cords about 8 weeks → germ cell nests about 10 weeks → primordial follicles with granulosa cells at 16–17 weeks → meiosis → primary follicles about 20 weeks. **Ovarian PGCs do enter meiosis; about 1 million at birth**.

- **Phenotypic sex** (internal and external genitalia, secondary sex characteristics) is also bipotential: every fetus has **wolffian and müllerian ducts**, a urogenital sinus, genital tubercle and urogenital swellings (Fig. 59.2); hormones determine the outcome.

TABLE 59.3 SELECTED GENETIC CAUSES OF 46,XY DIFFERENCES OF SEX DEVELOPMENT (DSDS)

Gene	Inheritance	Gonad	Uterus	External genitalia	Associated features
DISORDERS OF TESTIS DEVELOPMENT					
WT1	AD	Dysgenetic testis	+/-	Female or atypical	Wilms tumour , renal anomalies, gonadal tumours (WAGR, Denys-Drash, Frasier)
CBX2	AD	Ovary	+	Female	—
SF1	AR / AD (SL)	Dysgenetic testis / Leydig dysfunction	+/-	Female or atypical	Primary adrenal failure ; ovarian insufficiency in 46,XX relatives
SRY	Y	Dysgenetic testis or ovotestis	+/-	Female or atypical	—
SOX9	AD	Dysgenetic testis or ovotestis	+/-	Female or atypical	Campomelic dysplasia
MAP3K1	AD (SL)	Dysgenetic testis	+/-	Female or atypical	—
GATA4	AD	Dysgenetic testis	-	Female, atypical or male	Congenital heart disease
ZFPM2	AD	Dysgenetic testis	not stated	Female, atypical or male	—
ARX	X	Dysgenetic testis	-	Male or atypical	Developmental delay; X-linked lissencephaly
SAMD9	AD	Dysgenetic testis / Leydig dysfunction	+	Female, atypical or male	Myelodysplasia, infection, growth restriction, adrenal hypoplasia, enteropathy
DHH	AR	Dysgenetic testis / Leydig dysfunction	+	Female	Minifascicular neuropathy
MAMLD1	X	Dysgenetic testis / Leydig dysfunction	-	Hypospadias	—
DAX1	dup Xp21	Dysgenetic testis	+/-	Female or atypical	—
WNT4/ RSPO1	dup 1p35	Dysgenetic testis	+	Atypical	—
DISORDERS OF ANDROGEN SYNTHESIS (ALL AR; NORMAL TESTIS; NO UTERUS)					
LHR (LHCGR)	AR	Testis	-	Female, atypical or micropenis	Leydig cell hypoplasia
DHCR7	AR	Testis	-	Variable	Smith-Lemli-Opitz : coarse facies, 2nd-3rd toe syndactyly, failure to thrive, developmental delay, cardiac and visceral anomalies

Gene	Inheritance	Gonad	Uterus	External genitalia	Associated features
StAR	AR	Testis	–	Female or atypical	Congenital lipoid adrenal hyperplasia (primary adrenal failure)
CYP11A1	AR	Testis	–	Atypical	Primary adrenal failure
HSD3B2	AR	Testis	–	Atypical	CAH, adrenal failure ± salt loss; partial androgenisation from ↑ DHEA
CYP17	AR	Testis	–	Female or atypical	CAH; hypertension from ↑ corticosterone and 11-deoxycorticosterone (not in isolated 17,20-lyase deficiency)
CYB5A	AR	Testis	–	Atypical	Apparent isolated 17,20-lyase deficiency; methaemoglobinaemia
POR	AR	Testis	–	Atypical or male	Mixed 21-hydroxylase and 17 α -hydroxylase/17,20-lyase features; ± Antley-Bixler craniosynostosis
HSD17B3	AR	Testis	–	Female or atypical	Partial androgenisation at puberty; ↑ androstenedione:testosterone
SRD5A2	AR	Testis	–	Atypical or micropenis	Partial androgenisation at puberty; ↑ testosterone:DHT
AKR1C2 (AKR1C4)	AR	Testis	–	Female or atypical	Decreased fetal DHT production
DISORDER OF ANDROGEN ACTION					
Androgen receptor	X	Testis	–	Female, atypical, micropenis or normal male	Spectrum: CAIS (female) → PAIS (atypical) → normal male with infertility

AD, autosomal dominant; AR, autosomal recessive; SL, sex-limited; DHEA, dehydroepiandrosterone; DHT, dihydrotestosterone; WAGR, Wilms tumour, aniridia, genitourinary anomalies, intellectual disability; + present, – absent. Modified from Finlayson C, Jameson J, Achermann JC: Sex development. In Harrison's Principles of Internal Medicine, 21st ed, 2022, Table 390-4.

- **Pattern:** testis-development genes → **dysgenetic gonad, uterus often present** (Sertoli, hence antimüllerian hormone, also fail) and syndromic features. Synthesis and action genes → **normal testis, no uterus**; adrenal failure where the enzyme is shared with cortisol synthesis.
- **Trap:** a uterus in 46,XY points to dysgenesis, not to a synthesis or action defect. The androgen receptor is **X-linked**; synthesis defects are autosomal recessive.

Male-Typical Development

- Sertoli cells secrete **antimüllerian hormone (AMH) at 7–8 weeks** → müllerian regression **complete by about 10 weeks**.
- Leydig cells make **testosterone from about 9 weeks**: first autonomously, then under **placental human chorionic gonadotropin (hCG)**, later under fetal pituitary **luteinising hormone (LH)**. **Weeks 9–13: wolffian ducts → epididymis and vas deferens**, opening into the urogenital sinus at the müllerian tubercle.
- **5 α -reductase converts testosterone to dihydrotestosterone (DHT)**, which binds the androgen receptor with greater affinity and stability; in tissues with 5 α -reductase (**prostate, urogenital sinus, external genitalia**) **DHT is the active androgen**. External genitalia need systemic testosterone plus local DHT.
- Sequence (Fig. 59.3): tubercle elongates into the penis → urethral folds fuse **posterior to anterior** → prostate forms around the proximal urethra → urogenital swellings migrate posteriorly and fuse as scrotum → **complete by 12–13 weeks**. **Penile growth and testicular descent follow in the third trimester** under androgen and **insulin-like peptide 3 (INSL3)**.

Female-Typical Development

- Wolffian ducts regress, induced by **COUP-TF2** through inhibition of fibroblast growth factors. The ovary makes **no significant AMH**, so müllerian ducts form tubes, uterus and upper vagina.
- Vaginal origin is debated, but broadly **proximal two-thirds müllerian, distal one-third urogenital sinus**.
- **Without testosterone the external genitalia stay at the 6-week appearance**: tubercle → clitoris; lateral swellings → labia majora; urethral folds → labia minora, enclosing introitus and meatus (Fig. 59.5).

MALE-TYPICAL VERSUS FEMALE-TYPICAL DIFFERENTIATION

Structure	Male-typical (testis)	Female-typical (no testis)
Signal	AMH + testosterone → DHT	Absence of both
Wolffian ducts	Epididymis, vas, ejaculatory ducts, seminal vesicles	Regress
Müllerian ducts	Regress	Tubes, uterus, upper two-thirds of vagina
Urogenital sinus	Prostate, prostatic urethra, utricle	Distal one-third of vagina
Tubercle / folds / swellings	Penis / penile urethra / scrotum	Clitoris / labia minora / labia majora

From developmental steps to classification.

- Disruption of any step causes a DSD, and the 2006 classification mirrors the steps: abnormal karyotype → **sex chromosome DSD**; normal karyotype with abnormal gonadal development, or with abnormal hormone production or action → **46,XX or 46,XY DSD**.

Figures

- **Fig. 59.1 PGC migration** (De Felici, 2013): PGCs form about week 3 on the yolk-sac wall, pass through hindgut endoderm and dorsal mesentery, **reach the gonadal ridges by week 5**, and become oogonia or gonocytes over weeks 6–10.
- **Figs 59.2, 59.3, 59.5** (Martinez-Mora, 1994): undifferentiated genitalia (tubercle, urethral folds, genital ring, urogenital and anal membranes); male (glans, raphe, scrotum); female (clitoris, urethra, vagina, hymen, posterior commissure).
- **Fig. 59.4 Timetable** (White and Speiser, 2000; weeks approximate from the figure): Sertoli 6–7, Leydig 7–8, oogonia 9–10, oocytes 12, follicles 17, testicular descent from about 22. Wolffian derivatives from 8; female wolffian regression 10–14. Male müllerian regression 7–12; female müllerian fusion 9. Phallic ventral flexion 9; urethral closure and labioscrotal fusion 10–14; penile growth from 16; labia 10–18. Prostate 10, utricle 19; vagina reaches perineum 17.

KEY POINTS: TYPICAL SEX DEVELOPMENT

- Sex development occurs in three steps: chromosomal, gonadal and phenotypic sex.
- Chromosomal sex (46,XX, 46,XY or other) is established at fertilisation.
- **Gonads are bipotential until about 6 weeks; internal and external genitalia until about 8 weeks.**
- Where 5 α -reductase is present at differentiation (prostate, urogenital sinus, external genitalia), DHT is the active androgen.

Concept — *The testis imposes the male pathway through two separate hormones: AMH removes the müllerian ducts, testosterone (via DHT) builds wolffian and external structures. Without both, the female phenotype results — so any DSD phenotype can be reasoned from which hormone was missing, where and when.*

GENDER DEVELOPMENT

- **Gender identity:** identification of self — male, female or outside the binary. **Gender role (expression):** behaviour typical of male or female social roles. **Sexual orientation:** choice of sexual partner.
- Identity is shaped by biological, cognitive, social and cultural factors. Infants discriminate genders and form stereotypes in the **first year**; gender-typical behaviour appears **by 2 years**; **by 3 years** most children label themselves consistently.
- In DSD these inputs may differ: **identity different from designated sex is more common**, and some develop **gender dysphoria** (distress from that incongruence). **Avoiding dysphoria is a quality outcome**, which drives careful designation and has shifted designation trends; anyone may nonetheless affirm a different identity.

Concept — Designation at birth predicts an identity only the individual can later confirm; gender dysphoria is the outcome by which the prediction is judged.

DIFFERENCES OF SEX DEVELOPMENT: SPECIFIC CONDITIONS

- Classified **first by karyotype, then by mechanism** — gonadal development versus hormone production or action (Table 59.4).

TABLE 59.4 CLASSIFICATION OF DIFFERENCES OF SEX DEVELOPMENT (DSDS)

Category	Conditions
SEX CHROMOSOME DSD	
Karyotype	47,XXY (Klinefelter and variants); 45,X (Turner and variants); 45,X/46,XY (mixed gonadal dysgenesis) ; 46,XX/46,XY (chimerism/mosaicism)
46,XX DSD	
Gonadal (ovary) development	Gonadal dysgenesis; ovotesticular DSD; testicular DSD (e.g. SRY+, dup SOX9, RSPO1, NR5A1)
Androgen excess — fetal	3 β -hydroxysteroid dehydrogenase II (HSD3B2); 21-hydroxylase (CYP21A2) ; P450 oxidoreductase (POR); 11 β -hydroxylase (CYP11B1); glucocorticoid receptor mutations
— fetoplacental	Aromatase deficiency (CYP19); oxidoreductase deficiency (POR)
— maternal	Virilising tumours (e.g. luteomas); androgenic drugs
Other	Syndromic (e.g. cloacal anomalies); müllerian agenesis/hypoplasia (e.g. Mayer-Rokitansky-Küster-Hauser, MRKH); uterine anomalies (e.g. maturity-onset diabetes of the young type 5); vaginal atresia (e.g. McKusick-Kaufman); labial adhesions
46,XY DSD	
Gonadal (testis) development	Complete or partial gonadal dysgenesis (e.g. SRY, SOX9, SF1, WT1, DHH, GATA4/ZFPM2, MAP3K1); impaired fetal Leydig function (e.g. SF1/NR5A1, CXorf6/MAMLD1, HHAT, SAMD9); ovotesticular DSD; testis regression
Androgen biosynthesis	LH receptor (LHCGR); Smith-Lemli-Opitz; StAR; CYP11A1; HSD3B2; 17 α -hydroxylase/17,20-lyase (CYP17A1); POR; cytochrome b5 (CYB5A); 17 β -hydroxysteroid dehydrogenase III (HSD17B3); 5α-reductase II (SRD5A2) ; aldo-keto reductase 1C2 (AKR1C2)
Androgen action	Androgen insensitivity syndrome ; drugs and environmental modulators
Other	Syndromic male genital anomalies; persistent müllerian duct syndrome; vanishing testis; isolated hypospadias; congenital hypogonadotropic hypogonadism; cryptorchidism; environmental influences

Modified from Finlayson C, Jameson J, Achermann JC: Sex development. In Harrison's Principles of Internal Medicine, 21st ed, 2022, Table 390-1.

- **Pattern: 3 β -HSD, POR and OVT DSD appear under both karyotypes** — the enzyme defects virilise 46,XX and undervirilise 46,XY. **Trap:** 21-hydroxylase deficiency appears **only under 46,XX**; in 46,XY it causes no genital difference.

Sex Chromosome DSD

Klinefelter syndrome.

- **47,XXY — the commonest sex chromosome aneuploidy, 1 in 600 live-born males**; historically diagnosed in adolescence (incomplete puberty) or adulthood (infertility). Phenotype varies.
- Adolescent/adult features (Fig. 59.6) are summarised in the comparison table; neurodevelopmental differences may occur.
- **Diagnosis is moving earlier:** noninvasive prenatal testing (NIPT) and genetic testing for developmental delay. Most infants have no signs, but **micropenis, cryptorchidism, bifid scrotum and hypospadias** occur.
- **Testicular insufficiency is the hallmark.** Mini-puberty testosterone is low-normal to normal; measuring gonadotropins and testosterone then is recommended. Infant testosterone has not improved development (it may increase linear and penile growth), so it is **not currently recommended**.
- Puberty starts on time; insufficiency follows in adolescence or young adulthood as **seminiferous tubules hyalinise**, giving **firm testes under 3.5 cm**, high gonadotropins and low or low-normal testosterone. Most benefit from testosterone replacement. **Gynaecomastia reflects a raised oestradiol:testosterone ratio.**
- **Fertility:** most are azoospermic, but **testicular sperm extraction with intracytoplasmic sperm injection (ICSI)** achieves live births (meta-analysis). Warn of the high infertility rate; whether retrieval is better in adolescence is unclear, so **discuss once spermatarche is reached and the individual is ready**.
- Also: neurodevelopmental (cognitive, language, learning, attention, executive) difficulties needing therapy; low bone mineral density and raised fat mass; **extragonadal germ cell neoplasia and male breast cancer** (cumulative breast cancer risk by 75 years only about 1%); more autoimmune disease.

Turner Syndrome

- **Commonest female sex chromosome aneuploidy, 25–50 per 100,000 females:** one X with complete or partial loss of the second (some or all lines 45,X). Classic: **female genitalia, short stature, absent secondary sexual characteristics, cardiac and renal anomalies** (Fig. 59.7).
- **Diagnosis needs 45,X plus at least one typical manifestation** (short stature, hypergonadotropic hypogonadism, infertility, middle ear infection, malformations, other stigmata). **International guidelines 2017.** Increasingly prenatal (NIPT) rather than at short-stature or delayed-puberty evaluation.
- Karyotypes: **45,X 40–50%**; 45,X/46,XX 15–25%; 45,X/46,iso(X) 10%; **45,X/46,XY or other Y material 10–12%.** **Mosaicism → milder phenotype.**
- **Systemic features** (highly variable): endocrine (glucose intolerance, type 1 and 2 diabetes, hypothyroidism); gastrointestinal (raised liver enzymes, coeliac, inflammatory bowel disease); **coarctation and bicuspid aortic valve**; eye; ear (infection, hearing loss); webbed neck, neonatal peripheral oedema, short fourth metacarpal, hypoplastic nails, pigmented naevi; neurocognitive.
- **Renal anomalies in one-third to over half** (commonest in 45,X): horseshoe kidney, duplication or agenesis, malrotation; **multiple renal arteries** in most on cardiovascular imaging.
- Screen cardiac and renal at diagnosis, then lifelong (cardiovascular, autoimmune, eye, ear). **Growth hormone improves adult height.** **45,X/46,XY with atypical or male genitalia needs the same screening** but is not formally TS.

- **Ovarian insufficiency in about 95%** → oestrogen and progesterone. A minority have spontaneous thelarche, fewer menarche — **most likely with mosaicism, least with 45,X; TS+Y also more likely than 45,X**, with rare spontaneous live births. **AMH tracks karyotype, spontaneous puberty, gonadotropins and follicles.**
- **TS+Y:** Y material in an abnormal gonad → germ cell neoplasia; gonadoblastoma is benign but may progress to **germ cell cancer (GCC; dysgerminoma)**. **Neoplasia risk about 12–20%; GCC about 3%, in or after the second decade.**
- The 2017 guideline advises **gonadectomy at diagnosis**, but on low-level evidence, now challenged by weighing GCC against gonadal function and autonomy. The decision needs nuanced discussion but is **rarely urgent**. Gonadotropins and AMH gauge function; if gonads stay, **imaging is insufficient to detect GCC.**
- **Fertility:** spontaneous conception is rare, so most need assisted reproduction or fostering/adoption. **Oocyte donation** carries **higher pregnancy morbidity and mortality** in TS. **Oocyte cryopreservation** suits adequate adolescent ovarian function; ovarian tissue cryopreservation is investigational, notably at gonadectomy for TS+Y.

KLINFELTER VERSUS TURNER SYNDROME

Feature	Klinefelter	Turner
Stature	Tall	Short (growth hormone helps)
Phenotype	Small firm testes, gynaecomastia, sparse body and facial hair	Female genitalia, absent puberty, stigmata
Hormones	↑ gonadotropins, ↓ testosterone, ↑ oestradiol	Hypergonadotropic hypogonadism
Replacement	Testosterone	Oestrogen + progesterone
Fertility	Sperm extraction + ICSI	Oocyte donation; oocyte cryopreservation
Tumour	Extragenital germ cell; male breast	None without Y; TS+Y 12–20%

Mixed Gonadal Dysgenesis

- **Mixed gonadal dysgenesis (MGD): dysgenetic testis (variable position) + contralateral streak + persistent müllerian structures**, with variable androgen effect. Classic **asymmetry**: hypospadias, one palpable descended testis, one impalpable testis with a flat labioscrotal fold (Fig. 59.8).
- **Second commonest cause of neonatal atypical genitalia after CAH.** Usually **45,X/46,XY**, a mosaic seen in **1 in 5000 births**, not all of whom have MGD.
- 45,X/46,XY is a spectrum: female genitalia with variable ovaries (TS+Y), atypical genitalia (MGD), or **male-typical genitalia with normal testes (about 85%)**. **All carry Turner syndromic risk** (cardiac, growth).
- Presentation: variable clitorophallus, urogenital sinus or proximal hypospadias with labioscrotal fusion, undescended testis. **Hemiuterus, vagina and tube lie on the streak side; wolffian structures on the testis side** — the **local** action of AMH and testosterone.
- **Designation** is hard: marked androgen exposure predicts male identity more often; **dysphoria is commoner when raised female** (a systematic review found it in a substantial minority raised female, none raised male). Multidisciplinary assessment and transparent discussion precede designation; **irreversible infant surgery is discouraged unless medically necessary**.
- **Germ cell neoplasia 15–35%**, in streaks and dysgenetic testes alike → **remove streak gonads** (biopsy first if tissue type unknown).

- **Remaining gonad:** raised male — descended testis observed, **intra-abdominal testis brought down if possible**; raised female — remove, or consider relocation to inguinal canal or labioscrotal fold, preserving function should identity differ while allowing surveillance.
- At puberty the testis may make testosterone in any position (wanted or not); many fail, needing exogenous testosterone. **Infertility is typical**; sperm extraction has succeeded.

Ovotesticular DSD

- Ovarian and testicular tissue in one individual — **separate ovary and testis, or an ovotestis. Mostly 46,XX**; also mosaic or 46,XY; about **1 per 100,000**.
- Atypical, often asymmetric genitalia; **virtually all have a urogenital sinus, most a uterus**.
- An ovary sits in the normal position; a testis or ovotestis may lie anywhere on the path of descent. In a large South African cohort the **ovotestis was commonest** (about two-thirds of gonads), **ovaries more often left, testes right**.
- **Duct rule: a fallopian tube accompanies an ovary; a vas deferens always accompanies a testis.** An ovotestis has a tube in two-thirds and a vas ($\pm$ tube) in one-third.
- **Ovarian portion usually normal; testicular portion typically dysgenetic. Biopsy is often needed — polar biopsies** if components look distinct. **Neoplasia risk slightly raised (about 3%)**, in either component.
- Designation follows full assessment; historically based on predominant anatomy. Identity varies but often matches designation.
- Management is individualised; **irreversible procedures deferred** until the patient can participate. Removing discordant tissue is difficult even in polar ovotestes (Fig. 59.10); nonbinary rearing eases deferral to adolescence.
- Puberty follows the functioning tissue; the dysgenetic testicular part means **oestrogen effect often predominates**. Infertility is common, but births via ovum or sperm are reported.

HISTOLOGICAL PATTERNS OF THE OVOTESTIS

Pattern	Arrangement
Admixed	Central stromal core with mixed ovarian and testicular tissue
Compartmentalised	Ovarian upper pole; testicular lower pole within an ovarian mantle
Bipolar	Strict polar separation — suits polar biopsy

MIXED GONADAL DYSGENESIS VERSUS OVOTESTICULAR DSD

Feature	MGD	OVT DSD
Karyotype	45,X/46,XY	46,XX (also mosaic, 46,XY)
Gonads	Dysgenetic testis + streak	Ovary + testis, or ovotestis
Neoplasia	15–35%	About 3%
Gonadal step	Remove streaks	Polar biopsy; discordant tissue hard to separate
Systemic	Screen for Turner features	—

Figures

- **Fig. 59.6** Klinefelter: reduced beard and chest hair, narrow shoulders, gynaecomastia, wide hips, female pubic hair pattern, long limbs, hypogonadism. **Fig. 59.7** Turner: low hairline, low-set ears, small jaw, webbed neck, shield chest with wide-spaced nipples, cardiac defects, cubitus valgus, horseshoe kidney, short digits; short stature, failed puberty, infertility.
- **Fig. 59.8** MGD: clitorophallus with chordee, right dysgenetic testis in a formed hemiscrotum, perineal hypospadias. **Fig. 59.9** MGD: asymmetric anatomy; hemiuterus and opacified tube in a left inguinal hernia,

confirmed at exploration. **Fig. 59.10** Right polar ovotestis at 4 and 13 months: dysgenetic testis at one pole, well-developed ovary at the other.

KEY POINTS: SEX CHROMOSOME DSD

- Klinefelter: 47,XXY, small firm testes with insufficiency, gynaecomastia, tall stature, neurodevelopmental differences.
- Turner: female genitalia, short stature, ovarian insufficiency, renal and cardiac anomalies.
- Turner with Y: gonadal tumour risk; counsel on options including gonadectomy.
- MGD: 45,X/46,XY with asymmetric atypical genitalia; **second commonest cause of newborn atypical genitalia**.
- All 45,X/46,XY individuals need Turner screening.
- In MGD and OVT DSD, designation and surgical counselling are highly individualised.

Concept — Y material inside a dysgenetic gonad creates tumour risk; local rather than systemic hormone action creates asymmetry. TS without Y and KS fail symmetrically, are syndromic and carry no gonadal tumour excess; MGD and OVT DSD are asymmetric, each duct system following its own gonad.

46,XX DSD

- Arises from **abnormal gonadal development** or **androgen excess**.

Abnormal Gonadal Development

- **46,XX testicular DSD** (formerly sex reversal): testes and male genitalia in 46,XX, usually from **SRY translocated from Y to X**; about **1 per 20,000**. Normal fetal testosterone and AMH → **no müllerian structures**.
- **Azoospermia from loss of the azoospermia factor (AZF) region of Y**. Formerly found at infertility work-up; now by prenatal **genotype–phenotype discordance** (cell-free fetal DNA [cffDNA] 46,XX, ultrasound male). Progressive testicular regression and hypogonadism are common; identity typically male.

Abnormal Gonadal Development: Ovotesticular DSD

- **OVT DSD is most often 46,XX**, so it is also a 46,XX gonadal DSD (see Sex Chromosome DSD).

Excess Androgen Exposure: CAH

- **CAH is the commonest cause of 46,XX DSD**: reduced activity of one of **five cortisol-pathway enzymes** — side-chain cleavage, 3 β -HSD, 17 α -hydroxylase, 21-hydroxylase, 11 β -hydroxylase (Fig. 59.11).
- **21-hydroxylase deficiency (21-OHD): 95% of CAH**, autosomal recessive, about **1 in 15,000**. Cortisol deficiency → **excess adrenocorticotrophic hormone (ACTH)** → excess adrenal androgen; aldosterone deficiency → salt wasting.
- Salt-wasting severity follows residual enzyme activity and is **predictable by genotype**. Infants have variable virilisation with **bilateral nonpalpable gonads**, and risk **adrenal crisis (hyponatraemia, hyperkalaemia, hypotension, shock)** in the first weeks. **Raised 17-hydroxyprogesterone (17-OHP) is diagnostic**.
- **46,XX phenotype**: clitoromegaly to fused labioscrotal folds, graded by **Prader** (Fig. 59.12) — sometimes an apparently male infant (or distal hypospadias) with **impalpable gonads**. Ovaries are normal, so **müllerian structures are present**, often with a **common urogenital sinus** (vagina and urethra joining proximal to the perineum). Assignment is typically female; some with male designation and marked virilisation identify as male.

- **46,XY CAH** needs treatment but has no genital difference, so **newborn screening is most crucial for 46,XY**, who give no genital warning before crisis.
- Biochemistry: raised 17-OHP, androstenedione, testosterone and renin; low cortisol. **Measure 17-OHP after 36 hours** (normal neonatal surge). A **cosyntropin stimulation test** confirms cortisol adequacy more quickly.
- Salt wasting brings hyponatraemia, hyperkalaemia and metabolic acidosis, but **not usually before day 5–7** — **early normal electrolytes do not exclude CAH**. **Prompt glucocorticoid, mineralocorticoid and salt replacement is lifesaving**.
- **Long-term**: balance replacement and androgen suppression against **supraphysiological steroid harm** (growth attenuation, precocious puberty, obesity, hypertension, osteoporosis, cardiovascular disease). Hard because cortisol's circadian and stress responses are difficult to mimic and androgen also arises via the **"backdoor" pathway**; new drugs are in trials. **Classic CAH needs lifelong replacement**.
- **Fertility** is preserved but reduced. **46,XX**: fewer pregnancies, mainly because fewer seek motherhood, though ovulation may be affected. **46,XY: testicular adrenal rest tumours (TARTs)** — benign, commoner with poor suppression, **regress with more glucocorticoid**; they obstruct tubules (azoospermia, peritubular fibrosis, Sertoli and germ cell loss), so **sperm banking** is advocated; **excision rarely indicated**. Poor suppression can also cause **hypogonadotropic hypogonadism**.
- **Surgery**: feminising surgery (clitoroplasty, urogenital sinus repair/vaginoplasty) was historically done in infancy; now re-evaluated because it is **rarely urgent unless there is urinary retention** (technique Chapter 60).
- **21-OHD is a continuum** from salt-wasting classic to **nonclassic CAH (NC CAH)** (table below).
- **11 β -hydroxylase deficiency (about 5%)** and **3 β -HSD deficiency (under 1%, least common)** are both autosomal recessive and treated like 21-OHD; their distinguishing features are in the table. **3 β -HSD also undervirilises 46,XY**, because DHEA is weak.
- **Prenatal dexamethasone** largely prevents virilisation but **not the need for postnatal replacement**, and risks cognitive, cardiovascular and metabolic harm. Genital development is first-trimester, so it must start (by 6–7 weeks) **before fetal status is known**: number needed to treat **at least 4** even after NIPT excludes males — **three unaffected fetuses exposed for each affected girl helped**. The **Endocrine Society deems it experimental, only under an Institutional Review Board (IRB)-approved protocol**.

THE 21-HYDROXYLASE DEFICIENCY CONTINUUM

Form	Features
Classic salt-wasting	Cortisol and aldosterone deficient; crisis in first weeks; lifelong replacement
Simple virilising	No infant shock; cortisol deficiency with androgen excess
Nonclassic	Enzyme activity 20–50% ; about 1 in 200 in White populations (true prevalence unknown; diagnosis rising with prenatal testing); premature adrenarche, acne, hirsutism, advanced bone age, oligomenorrhoea; diagnosed any age; may need no treatment

FORMS OF CAH COMPARED

Enzyme	Share	Mineralocorticoid	BP / K ⁺	46,XX	46,XY	Signature
21-hydroxylase	95%	↓ salt wasting	Hypotension, hyperkalaemia	Virilised	Typical	↑ 17-OHP
11β-hydroxylase	About 5%	Excess	Hypertension, hypokalaemia	Virilised	Not listed as DSD	—
3β-HSD	Under 1%	↓ salt wasting	Hypotension, hyperkalaemia	Atypical (DHEA)	Undervirilised	↑ DHEA / DHEA sulfate

Enzyme	Share	Mineralocorticoid	BP / K ⁺	46,XX	46,XY	Signature
17α-hydroxylase / 17,20-lyase	—	Excess (corticosterone, 11-deoxycorticosterone)	Hypertension, hypokalaemia	—	Female or atypical; delayed puberty	↓ androgens

- **Pattern:** blocks before mineralocorticoid synthesis (21-OH, 3 β -HSD) waste salt; blocks that let 11-deoxycorticosterone accumulate (11 β -OH, 17 α -OH) cause hypertension and hypokalaemia; blocks upstream of androgens (3 β -HSD, 17 α -OH) undervirilise 46,XY. **Trap:** 17 α -hydroxylase deficiency causes **hypokalaemia**, not hyperkalaemia.

Excess Androgen Exposure: Maternal Androgens

- Rare transplacental androgen. **Fetoplacental aromatase (CYP19) deficiency** blocks conversion of fetal androgens to oestrogen, **virilising mother and infant**; the maternal effect resolves after delivery but **recurs in later pregnancies**. Also maternal adrenal tumours, luteomas, and synthetic progestogens or androgens.
- Exposure ends at birth, so **no postnatal medical treatment**; genital surgery decisions are multidisciplinary.

Figures

- **Fig. 59.11 Steroid pathway.** Cholesterol → (StAR, side-chain cleavage) → pregnenolone; 3 β -HSD converts each $\Delta 5$ steroid to $\Delta 4$. Mineralocorticoid: progesterone → 21-OH → deoxycorticosterone → 11 β -OH → corticosterone → 18-OH → 18-oxidase → aldosterone. Glucocorticoid: 17 α -OH → 17-OH progesterone → 21-OH → 11-deoxycortisol → 11 β -OH → cortisol. Sex steroid: 17,20-lyase → DHEA/androstenedione → 17 β -HSD → testosterone → 5 α -reductase → DHT; aromatase → oestrone/oestradiol.
- **Fig. 59.12 Prader (1958): grades I–V** of increasing masculinisation between normal female and male, in 46,XX CAH and applied to DSD generally.

KEY POINTS: 46,XX DSD

- CAH is most commonly due to 21-hydroxylase deficiency, causing atypical genitalia and salt wasting at around 1 week of life.
- Raised 17-OH progesterone is diagnostic of 21-hydroxylase deficiency.

Concept — *With an ovary there is no AMH, so müllerian structures are always present and virilisation must come from outside the gonad. The androgen source decides whether treatment is lifelong (CAH) or unnecessary (maternal); the level of the block decides whether salt wasting threatens life. 46,XX testicular DSD proves the rule: a testis, hence AMH, hence no müllerian structures.*

46,XY DSD

- Arises from **abnormal gonadal development** (complete or partial gonadal dysgenesis) or **abnormal androgen production or action**.

Complete and partial gonadal dysgenesis.

- **Complete gonadal dysgenesis (CGD, Swyer syndrome):** about **1 per 80,000**; presents as prenatal/infant discordance or **absent puberty**. Streaks have no hormonal or fertility function, so **prophylactic gonadectomy** is usual and hormone replacement lifelong; the **uterus grows with oestrogen and pregnancy is reported with ovum donation**.
- **Partial gonadal dysgenesis (PGD):** testis-promoting gene disruption (e.g. **WT1, SF1, SRY, SOX9, MAP3K1, DHX37**), often with extragenital features (WT1 → Denys–Drash and Frasier). Very rare; no incidence data. **Identity highly variable**, so designation needs discussion and surgery is individualised (as in MGD, OVT DSD). Pubertal testosterone is partial: males may still need testosterone, females need oestrogen.

COMPLETE VERSUS PARTIAL GONADAL DYSGENESIS (46,XY)

Feature	Complete (Swyer)	Partial
Gonad	Streak — no hormone, no gametes	Dysgenetic testis , variable testosterone
Genitalia	Female; müllerian structures present	Atypical external; variable internal
Identity	Typically female	Highly variable
Neoplasia	30–45%	Up to 60% if intra-abdominal
Gonad plan	Prophylactic gonadectomy	Individualised

Denys–Drash and Frasier syndromes.

- Both are **WT1** disorders combining gonadal dysgenesis and nephropathy. **Suspect Frasier in a female with steroid-resistant nephrotic syndrome, primary amenorrhoea and pubertal delay**; 46,XX Frasier presents with renal failure alone.

DENYS–DRASH VERSUS FRASIER SYNDROME

Feature	Denys–Drash	Frasier
Nephropathy	Congenital nephrotic syndrome (mesangial sclerosis); early end-stage renal disease	Focal segmental glomerulosclerosis; later, gradual
Wilms tumour	90%, often bilateral → high mortality → prophylactic bilateral nephrectomy	No known predisposition
46,XY genitalia	Atypical	Usually female (rarely male)
Gonadal tumour	About 40%	Gonadoblastoma, 60%

Absent (vanishing) testes syndrome.

- **Bilateral anorchia from testicular regression** (unilateral is frequent, bilateral very rare). **Timing sets the phenotype**: loss after the first trimester → male genitalia, impalpable testes, no müllerian structures; loss during it → **embryonic testicular regression** with atypical genitalia.
- Causes: often unknown — teratogen, **bilateral torsion**, genetic (**DHX37**; heterozygous **NR5A1** in a series with micropenis).
- **Mini-puberty (about 1–6 months): undetectable testosterone and AMH, raised follicle-stimulating hormone (FSH) and LH**. Testosterone replacement from adolescence; identity male; **prostheses** in later adolescence or adulthood.

Abnormal Hormone Production

- Synthesis defects may give atypical genitalia but **müllerian structures regress (AMH normal)**. Most are steroidogenic enzyme defects (Fig. 59.11) — **5 α -reductase deficiency** or rare salt-wasting CAH. Identity is variable; designation may need to be chosen at birth.

5 α -reductase deficiency.

- **Type 2 isoenzyme (SRD5A2) defect** → reduced testosterone-to-DHT conversion. At birth: female or atypical genitalia, urogenital sinus with labioscrotal fusion, **short blind vaginal pouch into which the vasa terminate**; testes and epididymides in labioscrotal folds, canals or abdomen.
- **Virilisation at puberty** (voice, muscle, phallic growth, erections); sperm production and fertility reported. Mechanism: high testosterone binding the receptor despite low affinity, or the pubertal rise in the **type 1 isoform**.
- **Testosterone:DHT above 20:1** — in mini-puberty, after hCG, or at puberty; **confirm by SRD5A2 sequencing**.
- Landmark: the **Dominican Republic "guevedoces" ("penis at 12")** cohort (Imperato-McGinley, 1979). **Many designated female later affirm male identity**; delay irreversible surgery.

Other synthesis defects.

- **17 β -HSD3 deficiency** (about **1 per 150,000**): one step above 5 α -reductase (androstenedione $\rightarrow$ testosterone); atypical at birth, **virilises at puberty**, with poor spermatogenesis and infertility.
- **LH receptor (LHCGR) variants: Leydig cell hypoplasia**, androgen deficiency.
- **3 β -HSD deficiency**: very rare CAH (**under 1 per 1,000,000**); DHEA accumulates, testosterone and DHT fall $\rightarrow$ **atypical genitalia/hypospadias in 46,XY**, plus salt wasting and adrenal crisis; atypical genitalia also in 46,XX.
- **17 α -hydroxylase deficiency** (usually with 17,20-lyase): low androgens with **hypertension and hypokalaemia** from excess corticosterone and 11-deoxycorticosterone; some female-typical, found at **delayed puberty**.

DEFECTS OF ANDROGEN SYNTHESIS IN 46,XY

Defect	Signature	Genitalia at birth	Puberty	Adrenal
LH receptor	No hCG response	Female, atypical, micropenis	—	Normal
17 α -hydroxylase	$\downarrow$ androgens; $\uparrow$ 11-deoxycorticosterone	Female or atypical	Delayed	Hypertension, hypokalaemia
3 β -HSD	$\uparrow$ DHEA	Atypical / hypospadias	—	Salt wasting
17 β -HSD3	$\uparrow$ androstenedione:testosterone	Atypical	Virilises	Normal
5 α -reductase 2	Testosterone:DHT above 20:1	Female or atypical	Marked virilisation	Normal

- **Pattern:** AMH is normal in all, so **no uterus**; adrenal disease only where the enzyme is shared with cortisol synthesis; the two most downstream defects are those that **virilise at puberty**.

Abnormal hormone action.

- **Androgen receptor variants (X-linked)** cause complete or partial androgen insensitivity (CAIS, PAIS).
- **CAIS:** female genitalia, **inguinal or abdominal testes, short vagina, no müllerian structures**; X-linked recessive, 1 in 20,000–90,000 46,XY births. Historically found at **primary amenorrhoea** work-up or when testes appeared at herniorrhaphy. Testosterone is made but not sensed; **aromatisation produces breasts at puberty**.
- **About 50% of CAIS have an inguinal hernia; 1–2% of girls with inguinal hernia are 46,XY CAIS** $\rightarrow$ at female herniorrhaphy, **vaginoscopy for a cervix or endoscopy through the sac for an intra-abdominal testis**. Prenatal diagnosis by NIPT-ultrasound discordance is increasing.
- Histology: absent or incomplete spermatogenesis, normal or hyperplastic Leydig cells — like immature cryptorchid testes.
- Early prophylactic gonadectomy was once standard, but **prepubertal tumour risk is about 2%**, so **gonads stay until after breast development** and the patient can decide; some report better quality of life on endogenous hormones. No gametes; a gonadal management protocol has been proposed. **Dilation usually suffices; vaginoplasty rarely needed**.
- **PAIS:** residual receptor function; **more than 800 variants**; about **1 per 100,000**. At puberty some androgen effect and often some breast growth. Identity variable, **most often male**. Raised male: high-dose testosterone may improve genital development. Female identity: **gonadotropin-releasing hormone (GnRH) agonists** or gonadectomy, then oestrogen. **Neoplasia about 30% for undescended gonads** (Table 59.7: 50% non-scrotal). Fertility reduced but possible; care individualised as in MGD and OVT DSD.

COMPLETE VERSUS PARTIAL ANDROGEN INSENSITIVITY

Feature	CAIS	PAIS
Genitalia	Female; short vagina	Spectrum of atypical
Identity	Female	Variable, mostly male
Tumour risk	About 2% prepubertal	About 30% undescended
Gonad plan	Keep until after breast development	Careful discussion
Fertility	None via sperm	Reduced, possible

Persistent müllerian duct syndrome.

- **PMDS (hernia uteri inguinale): 46,XY, male genitalia, testes with vasa and epididymides, plus müllerian structures** — undescended testes, bilateral tubes, uterus and upper vagina draining into the **prostatic utricle**; usually found at herniorrhaphy or orchidopexy.
- **Autosomal recessive: AMH (19p13) or AMH type II receptor (12q13)**, one of which is mutated in the large majority. Three presentations (Clarnette; Fig. 59.13) in the table; **PMDS underlies 30–50% of transverse testicular ectopia**.
- Sequelae: hernia, infertility (abnormal testes, abnormal vasa/epididymides, surgical injury), testicular tumours (**seminoma**, reflecting intra-abdominal cryptorchidism) and **müllerian malignancy**. Clues: consanguinity, familial infertility or undescended testes.
- **Orchidopexy is recommended**. Because the **vasa hug the uterus and proximal vagina**, remnants were historically preserved (divided longitudinally if needed to reach the scrotum). **Remnant malignancy 3–8%** → discuss removal versus ultrasound or MRI surveillance; if removing, **laparoscopic or robotic** access spares the vasa.

CLARNETTE CLASSIFICATION OF PMDS

Category	Testis position	Share
1	Bilateral intra-abdominal , in ovarian position	60–70%
2 (classic)	One testis in hernia sac or scrotum + contralateral inguinal hernia	20–30%
3	Both in one sac (transverse testicular ectopia) with tubes and uterus	10%

KEY POINTS: 46,XY DSD

- Complete gonadal dysgenesis carries a 30–45% gonadal tumour risk.
- Denys–Drash: atypical genitalia, congenital nephrotic syndrome, Wilms tumour.
- 5 α -reductase deficiency: female or atypical genitalia at birth, virilisation at puberty.
- **Strongly consider leaving the gonads in complete androgen insensitivity in situ until after puberty.**
- PAIS shares the clinical challenges of MGD and OVT DSD.

Concept — In 46,XY DSD the uterus separates the mechanisms: *dysgenetic testes fail at AMH and testosterone (uterus present); synthesis and action defects keep AMH (no uterus); PMDS is the mirror image — AMH failure alone with normal virilisation. Tumour risk tracks dysgenesis and intra-abdominal position, which is why streaks come out and CAIS gonads can wait.*

Other Anatomic Conditions

- Also managed by DSD teams: **aphallia, vaginal duplication and other vaginal anomalies, cloacal exstrophy (Chapter 43), micropenis and Mayer–Rokitansky–Küster–Hauser (MRKH) syndrome.**

MRKH syndrome.

- **Congenital absence of uterus and vagina**, mostly sporadic, **1 in 4000–5000 female births**, 46,XX; female external genitalia and secondary sex characteristics with a **shallow vaginal pouch** (Fig. 59.14: T2 MRI, no uterus or vagina between bladder and rectum).
- **Presents mainly with primary amenorrhoea**, also infertility or dyspareunia. **Upper tract anomalies in about one-third** (agenesis, pelvic kidney, horseshoe); cardiac anomalies in a minority.
- Many loci, often none found: 1q21.1 (RBM8A), 1p31–1p35 (WNT4), 7p15.3 (HOXA), 16p11 (TBX6), 17q12 (LHX1, HNF1B), 22q11.21, Xp22. Comorbidity predicts the gene: **RBM8A → TAR syndrome** (thrombocytopenia, absent radius); **only WNT4 → raised androgens**.

TYPICAL VERSUS ATYPICAL MRKH

Feature	Typical	Atypical (up to 10%)
Anatomy	Symmetrical uterine remnants; normal tubes, ovaries and ovarian function	Asymmetrical remnants and/or tubal aplasia
Endometrium	—	Haematometra → cyclical pain
Upper tract anomalies	Absent in the reported series	Much more common

VAGINAL CREATION IN MRKH

Option	Notes
Dilation	First-line (American College of Obstetricians and Gynecologists); outcomes comparable to vaginoplasty
Vecchietti	If dilation impossible: progressive traction on the vault by a device at the vaginal dimple; short minimally invasive procedure
Skin flap	Former first-line surgery; no lubrication, introital scarring
Bowel (colon, small bowel)	Lubricated, available; out of favour — overstretching, excess mucus, better alternatives
Buccal mucosa	Partial or complete; primary perineal approach; major improvement
Tissue-engineered	Off-the-shelf grafts (e.g. small intestinal submucosa) avoid harvest; emerging

- **Uterine remnants**: if symptomatic, remove or suppress menses hormonally; **never connect to a neovagina — there is no cervix**.
- **Uterine transplantation**: landmark first trial (**Brännström**) — **nine** recipients, **two** early graft removals, **multiple live births**; still investigational, with trials in several countries.

Micropenis.

- **Stretched penile length more than 2.5 SD below the mean — under 2 cm at 0–5 months**. A true micropenis (not buried) **needs endocrine evaluation** for DSD or **central hypopituitarism** (cortisol, growth hormone deficiency). Money-era female assignment is **no longer recommended**.

Concept — *In MRKH the ovaries are normal, so management targets function, never the gonad — a vagina by dilation first, fertility by transplantation. Micropenis is a marker of endocrine disease, not a reason to reassign sex.*

MULTIDISCIPLINARY (INTERDISCIPLINARY) CARE MODEL

- The model changed because **long-term outcomes under historical care were suboptimal**. That care followed **John Money (1975)** — early designation and strongly advised early surgery (table). Intersex activists and clinicians challenged it, and it is now judged **ill-advised and against psychosocial interests**.

- **Lifelong psychosocial care** is needed: anxiety, depression, suicidal ideation and related conditions are more common. Care is shifting towards **bodily autonomy and shared decision-making**.
- The **2006 consensus made multidisciplinary care the standard**; the 2016 update stressed teamwork and **peer support groups**. Teams should work **interdisciplinarily** — as one, not a sum of parts.
- No "one size fits all" algorithm: care is individualised to patient, family, culture and social setting.
- **Team**: endocrinologist; urologist, paediatric surgeon, gynaecologist; geneticist and genetic counsellor; psychologist, psychiatrist, social worker; ethicist; nursing and support staff; ideally **individuals with DSD/intersex traits** as educators and peer links.
- Evidence is often lacking, so the team is **transparent about what is known and unknown**; without consensus it **educates and supports the family's decision rather than dictating it**.
- **A urologist without a team should recognise this and transfer the patient to an appropriate centre as soon as possible, even a newborn**.
- Despite cost pressures (lobby institutions and legislators), **clinicians should see the patient together in one clinic — separate days are not recommended** — so information is exchanged in real time and a shared plan agreed. The same team sees inpatients; **urgent (within 24 hours) review** usually arises in the **neonatal intensive care unit (NICU)** for atypical genitalia.
- **Transition** is harder than in most urology (complexity, rarity, many specialists); adults may be seen in paediatric units or by paediatric specialists in adult settings. Patients must not be "abandoned": needs persist or grow, especially **fertility and sexual health**, which are hard to raise. Needs include coordination, autonomy, open discussion and psychosocial care; long-term outcomes remain to be defined, so **continuity must span life**.

HISTORICAL VERSUS CURRENT MODEL

Element	Historical (Money)	Current
Team	Endocrinologist + one surgeon	Interdisciplinary, same-day clinic , peer support
Decision-maker	Surgeon (designation and surgery)	Family and patient , after education
Disclosure	Not encouraged (secrecy)	Transparent
Surgery	Early; presented as optimal	Individualised; irreversible steps deferred
Psychosocial care	Often absent	Integral, lifelong

KEY POINT: MULTIDISCIPLINARY (INTERDISCIPLINARY) CARE MODEL

- An interdisciplinary model in which clinicians see patients together on the same day is recommended.

Concept — Money's premise was that gender follows early designation and surgery. Its failure moved decisions **from surgeon to family and ultimately patient**, and the team's role **from direction to education**; co-location, transparency and lifelong continuity follow.

INFANT PRESENTATION AND EVALUATION

- Classically **atypical genitalia at birth**; increasingly **prenatal** through cffDNA — atypical genitalia on ultrasound, sex chromosome aneuploidy or mosaicism, or **genotype-phenotype discordance** (e.g. Y material with female genitalia at the 20-week scan).
- **Prompt multidisciplinary medical and psychosocial evaluation, ideally in the first 24–48 hours**; atypical genitalia may mean an emergency (**salt-wasting CAH**).

- **Communication: congratulate**; use "your baby", they/them or the baby's name — **never "it"**; train perinatal staff, since inconsistency breeds distrust; the NICU itself can frighten families. Apart from CAH **most DSDs are not life-threatening**, so reassure that **designation and treatment are not emergent**.
- **Evaluation** covers prenatal history, comorbidities, anatomy, chromosomes and hormones, guiding testing, parental education and (often) designation. **Family history**: infertility, recurrent pregnancy loss, amenorrhoea, hirsutism, relatives with DSD (e.g. CAIS, hypospadias) or atypical genitalia; **infant death suggests CAH**. Ask about maternal steroids or contraceptives.
- **Examination** in gender-neutral terms (Fig. 59.15), measuring **stretched clitorophallic length, glans width, urethral plate length and anogenital distance**, and locating the meatus versus a single perineal opening.
- **Gonads are the most informative sign: bilateral impalpable gonads with atypical genitalia → CAH; any palpable gonad means testicular tissue**, excluding all but the rarest CAH, because **ovaries do not descend**. Rarely an ovotestis reaches the canal, suspected from asymmetric pole texture and ultrasound.
- **Bilateral impalpable testes, or unilateral impalpable testis with hypospadias = DSD until proven otherwise**. Landmark **Kaefer (1999)**, cryptorchidism + hypospadias without atypical genitalia: **DSD 30% overall; about 15% if the testis was palpable, 50% if impalpable; proximal meatus 65% versus 5–8% midshaft/distal**. Later work supports testing proximal hypospadias and links **penoscrotal transposition and bifid scrotum** to a DSD diagnosis.
- **Anogenital distance** reflects fetal androgen (correlating with testis size, sperm count, penile length and testosterone in rats and humans) but **lacks normative values**. The **External Masculinisation Score** (phallus size, labioscrotal fusion, gonad and meatal position) has **unproven predictive value**.
- **Ultrasound** (pelvic/scrotal): the **neonatal period is ideal for the uterus** (maternal hormones enlarge it), so **routine rectal examination is unnecessary**. It finds abdominal gonads and suggests type, but **normal versus dysgenetic testis is hard to tell and streaks are often invisible**. **Asymmetry suggests MGD if 45,X/46,XY, OVT DSD if 46,XX**.
- **Karyotype** at once: **at least 45 cells**, laboratory warned of DSD; preliminary in **2–3 days**. **Fluorescence in situ hybridisation (FISH)** identifies X and Y **within hours**, usually to confirm a second X.
- **Endocrine timing matters**. If CAH is possible, **17-OHP at about 36 hours, before discharge**. At about **1 week**: LH, FSH, testosterone, oestrogen, androstenedione, DHT and AMH (Table 59.5).

TABLE 59.5 ENDOCRINE TESTING FOR DSD EVALUATION

Test	Interpretation
17-hydroxyprogesterone	Raised in CAH
Antimüllerian hormone	Normal → functioning Sertoli (testis) or granulosa (ovary) cells; also ovarian reserve
LH, FSH	Raised → poor gonadal function
Testosterone / oestrogen	Whether functioning tissue is testicular or ovarian
Androstenedione	Raised in 21-OHD and 17β-HSD3 deficiency
DHEA sulfate	Raised in 21-OHD and 3β-HSD deficiency
Testosterone:DHT	Above about 20:1 → 5α-reductase deficiency
hCG stimulation	No testosterone rise → no testis, no Leydig cells, or LH receptor defect

- **Pattern**: tests localise the gonad (AMH, sex steroids, gonadotropins, hCG), the adrenal enzyme (17-OHP, androstenedione, DHEA sulfate) or peripheral conversion (testosterone:DHT). **Trap**: androstenedione and DHEA sulfate are each shared by two defects; hCG non-response has three causes.
- **Gonad size and position and presence of a uterus narrow the differential** (Figs 59.16 and 59.17, below). **Precise diagnosis is nearly always reached in 46,XX DSD but in under 50% of 46,XY DSD**.

- If unclear: **MRI, diagnostic laparoscopy ± gonadal biopsy, genitography, endoscopy with examination under anaesthesia** (Fig. 59.18). **No gonadectomy or removal of reproductive organs during the diagnostic phase.** Genetic tests (Table 59.6), now cheaper and faster, are tailored to suspicion and comorbidities.

FIG. 59.16 ALGORITHM: FEMALE-TYPICAL GENITALIA WITH Y MATERIAL ON PRENATAL TESTING

Karyotype	Examination and ultrasound	Likely diagnosis
46,XY	Testes (abdominal or palpable); no uterus	CAIS
46,XY	Streak/impalpable gonads; uterus	Complete gonadal dysgenesis
45,X/46,XY	Streak/impalpable gonads; uterus	Turner + Y

FIG. 59.17 ALGORITHM: ATYPICAL GENITALIA

Karyotype	Examination, ultrasound, endocrine	Likely diagnosis
46,XX	Ovaries/impalpable; uterus; ↑ 17-OHP, ↑ testosterone	CAH
46,XX	Gonads not two ovaries; uterus ±	OVT DSD; rare causes
46,XY	Variable gonads; uterus ±	Partial gonadal dysgenesis; OVT DSD
46,XY	Variable gonads; no uterus	Androgen synthesis or action defect
45,X/46,XY	Variable gonads; uterus ±	MGD; OVT DSD
Palpable gonads: 0 → CAH; 1 → PGD, PAIS, MGD, OVT DSD; 2 → 46,XY DSD (PAIS, 5α-reductase deficiency, Leydig cell hypoplasia, PGD, rare CAH)		

TABLE 59.6 ADJUNCTIVE GENETIC TESTING OPTIONS

Test	Indication
FISH for SRY	Genitalia–karyotype discrepancy (e.g. 46,XX with masculinised genitalia)
Single gene	Confirm a strongly suspected diagnosis
DSD gene panel ("DSD slice")	Known DSD-associated variants
Microarray	Rapid scan (including new candidate genes) when diagnosis unclear and/or multiple anomalies
Whole exome	As microarray; may diagnose when others fail
Whole genome	Could replace exome, panels and microarray; new, not universally available

Modified from Granberg CF, editor; Yerkes E, Johnson EK: Disorders of sexual differentiation. AUA Core Curriculum, 2023.

Figures

- **Fig. 59.15** Seven reflections of androgen exposure to record: clitorophallic length; glans width; number and position of perineal orifices; labioscrotal fusion; labioscrotal pigmentation and rugation; anogenital distance; location of palpable gonads — using gender-neutral terms until shared decisions on sex of rearing are possible.
- **Fig. 59.18** Laparoscopy in a 1-year-old with MGD: left streak gonad; right dysgenetic testis and uterus.

KEY POINTS: INFANT PRESENTATION AND EVALUATION

- Prenatal suspicion of DSD is increasing, largely through cell-free fetal DNA.
- Initial evaluation: multidisciplinary team, sensitive gender-neutral terminology, congratulations.
- Bilateral impalpable gonads may indicate CAH; asymmetry suggests OVT DSD or MGD.

- **No gonadectomy or removal of reproductive organs in the diagnostic phase.**

Concept — *Neonatal work-up is a triad — karyotype, gonads (number and position), uterus — with 17-OHP the one test that cannot wait; everything irreversible is deliberately deferred while the triad is assembled.*

OLDER CHILD/ADOLESCENT PRESENTATION AND EVALUATION

- Classic presentation at expected puberty: **primary amenorrhoea, absent puberty, or virilisation**; also via genetic testing for short stature or developmental concerns. **Late presenters have typical external genitalia**: CGD, KS, CAIS, MRKH, TS+Y — and 5 α -reductase deficiency, female at birth, virilising at puberty.
- **Basal hormones without hCG are more revealing than in infancy: high gonadotropins with low sex steroid → primary gonadal failure; high LH and testosterone with female genitalia → CAIS.**
- **Examine with patience** — adults recall childhood genital examinations negatively, some with post-traumatic stress symptoms. Limit clinicians present; examine after the first visit; **examine only when management may change**; use examination under anaesthesia selectively.

Concept — *The adolescent axis is awake, so basal levels answer what infancy needed hCG or mini-puberty to show; late presenters are exactly those whose genitalia look typical.*

PEER SUPPORT

- Peer support offers **emotional and informational support through shared experience**. Evidence is limited and **referral and engagement are low**; success depends on meeting real needs and **training facilitators**. Some prefer broad groups, others **diagnosis-specific** ones.
- Members value **"not feeling like the only one"**, navigating systems, finding specialists and learning; it is empowering but emotionally taxing. **Advocacy** (public perception, medical care, legislation) suits some and deters others.

CURRENT "HOT TOPICS" AND CONTROVERSIES

Sex Designation and Future Gender Identity

- **No one's future gender identity can be predicted with certainty**. Identity differing from assigned sex is **more common than once recognised**, even after diligent diagnosis; a small minority transition, though rates differ by diagnosis.
- **Biology**: lower prenatal androgen exposure → female identity, higher → male; **no fetal androgen (CAIS, CGD) → female identity**. Postnatal androgens may add, via the brain and via appearance, self-image and societal response. Females with CAH show more male-typical roles; 5 α -reductase deficiency more often male.
- **Yet sex of rearing predicts satisfaction with gender identity better than androgen exposure**. Designation weighs probable adult identity, sexual function, hormone and fertility potential, fetal androgen exposure and psychosocial factors.

GENDER DYSPHORIA BY DIAGNOSIS AND REARING (SYSTEMATIC REVIEW BOTTOM LINE)

Diagnosis	Rearing	Dysphoria
46,XX CAH	Female	Lowest
PAIS	Either	Occurs with both; more when raised male
MGD	Female	Substantial minority
5 α -reductase deficiency	Female	Highest, approaching half

CURRENT DESIGNATION TRENDS

Female	Male (46,XY, more androgen exposure)
46,XX CAH; CAIS; CGD; 46,XY LH receptor deficiency	PGD; PAIS; 5 α -reductase deficiency; 17 β -HSD3 deficiency

- **Outcome data are unreliable:** evolving concepts of sex and gender change reporting; greater acceptance of intersex, transgender and nonbinary identities means **older studies may understate dysphoria**; and the recent **trend towards male assignment** has not yet reached adulthood. Updated studies are needed.
- **The "social emergency" framing is outdated.** The team should **meet the family in the first 1–2 days** to start work-up and set expectations on timing of designation. A **psychosocial member is critical**, explaining the biological complexity and frequency of variation.
- **There is no test for gender identity** except waiting for the individual's own sense of it. The team supports the family's choice (male, female or other, and timing) and **lets the family lead on terms and pronouns**.

Concept — Designation is a probabilistic prediction from androgen exposure, but rearing and culture weigh as heavily; the aim is an informed choice made without false urgency, keeping options open until the child can speak for themselves.

Role and Timing of External Genital Surgery

- **The most argued controversy.** Elective surgery before the age of consent may irreversibly create anatomy incongruent with later identity and bypasses autonomy; against this stand possible functional advantages and the psychosocial burden of unoperated atypical genitalia in a binary society. "Feminising" and "masculinising" are contested terms, used for want of alternatives.
- **Feminising genitoplasty = clitoroplasty, vaginoplasty and labioplasty**, staged or combined (Chapter 60). Techniques and outcomes have improved; **scrutiny is of timing and who decides**.
- Claimed benefits of early surgery (better tolerance, no memory, easier technique, fewer complications) are **unproven by large studies**, and Money's rationale is **incorrect**. For those whose identity later differs, **early clitoral reduction can be catastrophic**; others regret complications or exclusion from the decision.
- Surveys nonetheless show **parents and patients mostly prefer early surgery**, including most 46,XX adults after feminising genitoplasty (systematic review) — limited by **no control group** and inclusion of unoperated patients.
- **Recommendation:** transparent multidisciplinary discussion of early versus later or no surgery; shared decision-making (rarely formalised); ideally **no irreversible genital surgery until the individual can participate**. Parental consent must address the **limits of predicting identity, embodiment goals and fertility wishes**.
- **CAH:** most data come from CAH, and some groups (**CARES Foundation**) argue CAH is not DSD and early-surgery concerns do not apply. But **dysphoria is increased in CAH**, some thrive raised male, and not all women want or would risk genital surgery — so infant feminising surgery is questioned even after full counselling. Whether CAH should be treated separately is unresolved.
- **Masculinising surgery is mostly proximal hypospadias repair**, often with **penoscrotal transposition**. It retains a **high complication rate** even in high-volume single- and multistage series, with concern for **erectile dysfunction after multiple operations**. Whether a defined DSD raises complications is disputed: one series found higher rates; a **prospective multi-institutional 46,XY DSD study found them similar**.

FEMINISING VERSUS MASCULINISING SURGERY

Feature	Feminising	Masculinising
Reversibility	Irreversible (clitoral reduction)	Mostly not irreversible
Case for infancy	Unproven; surveys favour early	Healing, fewer complications, less memory; wish to void standing at school age

Feature	Feminising	Masculinising
Functional need in infancy	Rare (urinary retention)	None — elective
Current stance	Defer until the individual can participate	Infant repair often chosen when raised male, after full counselling

Concept — The timing debate turns on reversibility: *clitoral surgery cannot be undone, so the decision moves towards the patient's own consent; masculinising repair is largely revisable, so infant surgery remains common when a family raises the child male.*

Gonadal Management: A Paradigm Shift

- From **automatic gonadectomy** (dysgenetic or discordant gonads removed early as nonfunctional and tumour-prone) to **individualised assessment of tumour risk, function and autonomous choice**. Risks by diagnosis and age are now **lower than once stated** and stratified (Table 59.7); tumour biology makes **biopsy** increasingly decisive.
- **Fertility is being reconsidered**: future gametes may allow biological fertility even if discordant with identity; **germ cells were found in DSD gonads removed as "nonfunctional"**; in vitro follicle growth and sperm maturation from immature tissue are investigational. Some centres offer **experimental gonadal tissue cryopreservation** at prophylactic gonadectomy.
- Genetic diagnosis will personalise fertility and tumour estimates. Alternatives to removal include **initial biopsy** or **moving an at-risk intra-abdominal gonad to the inguinal canal or labioscrotal fold**.

TABLE 59.7 GERM CELL TUMOUR RISK AND RECOMMENDATIONS FOR MANAGEMENT IN DSD

Risk	DSD	Risk	Management
High	Gonadal dysgenesis ^a (+Y) ^b , intra-abdominal	15–35%	Gonadectomy ^c
	Frasier syndrome	60%	Gonadectomy ^c
	Denys-Drash (+Y) ^b	40%	Gonadectomy ^c
	PAIS, non-scrotal gonad	50%	Gonadectomy ^c
Intermediate	Turner (+Y) ^b	12%	Gonadectomy ^c
	17β-HSD3 deficiency	28%	Monitor or gonadectomy (uncertain)
	PAIS, scrotal gonad	Unknown	Biopsy ^d and irradiation
	Gonadal dysgenesis (+Y) ^b , scrotal gonad	Unknown	Biopsy ^d and irradiation
Low	CAIS	2%	Biopsy ^d ; next step undetermined
	OVT DSD	3%	Remove testicular tissue (uncertain)
None (?)	5α-reductase deficiency; Leydig cell hypoplasia	Unknown	Unresolved

^a Includes not otherwise specified, 46,XY, 45,X/46,XY, mixed, partial, complete. ^b GBY (gonadoblastoma locus on Y) region positive, including TSPY. ^c At diagnosis. ^d At puberty, examining **at least 30 seminiferous tubules**, preferably with **OCT3/4 immunohistochemistry**. Modified from Looijenga LH et al.: Tumor risk in disorders of sex development. Best Pract Res Clin Endocrinol Metab 21:480–495, 2007, Table 3.

TUMOUR RISK: BODY TEXT VERSUS TABLE 59.7

Condition	Text	Table 59.7
Complete gonadal dysgenesis	30–45%	15–35%
Partial gonadal dysgenesis	Up to 60% intra-abdominal	15–35%
Turner + Y	12–20% (GCC about 3%)	12%

Condition	Text	Table 59.7
PAIS	About 30% undescended	50% non-scrotal

- **Why they differ:** the text uses later diagnosis-specific series; the table is a 2007 pooled compilation grouping all gonadal dysgenesis. (MGD 15–35%, Frasier, Denys–Drash, CAIS and OVT figures agree.)
- **Pattern:** risk rises with **Y material (GBY/TSPY)**, **dysgenesis and intra-abdominal position**, falls with a scrotal gonad; high → gonadectomy at diagnosis, intermediate → pubertal biopsy, low → preserve. **Trap:** CAIS is **low risk** despite abdominal testes; Frasier carries the table's **highest** figure despite no Wilms risk.

Concept — Gonadal tumour risk is set by Y material, dysgenesis and intra-abdominal position, so management has moved from "remove every discordant gonad" to "quantify risk, preserve function and fertility where risk allows, and wait for the patient".

SUMMARY AND CONCLUSIONS

- Traditional DSD care proved suboptimal and recommendations are evolving. **The urologist's role belongs only within a multidisciplinary model.** Care is individualised; the **role and timing of surgery are being redefined**; counselling must be **transparent about the known and unknown**. Long-term outcomes with and without surgery, and genetic insight, will improve care.